Fundamental research

Binxu Jiang
Student number - 1524313

Siqi Ye
Student number - 0316377

Sander Snoeck
Student number - 2129930

Carolina Aranda Bassegoda
Student number - 9809643

1 Introduction

The following assignment which was designed by Marc J. van Kreveld is inspired in his work designing and analysing the Gourds puzzle along with Joep Hamersma, Yushi Uno, and Tom C. van der Zanden [1].

The puzzle discussed here consists of the reconfiguration of a board covered in dominoes except for two squares. The players can move the dominoes via four kinds of methods including straight slide, 90-degree turn, sideways slide, and pivot. This puzzle aims to reach the goal configuration from the start configuration by moving dominoes. In order to discuss the tractability of this puzzle we will begin by defining the components of this puzzle formally.

2 Preliminaries

Definition 2.1. *A valid **Board**, B, consists of a square grid with* $2n + 2$ *squares where n is the number of Domino pieces in the puzzle. A board must have no holes and be 2-connected, meaning that if any square is removed from the board, it does not split the board into two components.*

Definition 2.2. *The **dual graph**,* G_B*, of a board is such that each vertex in* G_B *corresponds to the center of a square in B and the edges of* G_B *connect the vertices corresponding to adjacent squares in B.*

Definition 2.3. *A **Hamiltonian cycle** (of a board), H, is a path that visits every vertex of* G_B *exactly once. Furthermore, the starting and finishing nodes must be the same.*

Definition 2.4. *Nothing is a **Valid move** unless generated by finite repetitions of the following moves:*

i) *Straight slide. Moving the domino one square in the direction of the domino.*
ii) 90° *turn. If there is an empty square adjacent to a part of a domino, the part of the domino adjacent to the empty square can occupy the empty square, and the other part of the domino occupies the position of the part that is moving to the empty square in one* 90° *turn.*
iii) *Sideways slide. If the two squares adjacent to a domino are both empty, the domino can sideways slide to occupy the two empty squares.*
iv) *Pivot. If the two squares adjacent to a domino are both empty, one part of the domino can be pivoted in* 90° *around another part of the domino to occupy one empty square.*

Remark 2.5. Let G be a graph corresponding to a connected square grid (which admits a Hamiltonian cycle) and H its corresponding Hamiltonian cycle, where H has at least one vertex, v_f, corresponding to an empty square. A domino occupying the square corresponding to the node adjacent to v_f in H can always be moved into the free square using a 90° *turn* or a *Straight slide* as in Definition 2.4. This results in the domino occupying the square that was free and the square corresponding to the vertex adjacent to v_f in H.

3 Puzzle tractability

In this section, we will show that any two configurations of the board can be transformed into each other. In order to do this we will begin by proving five lemmas which we will use in the proof of our theorem. The proofs in this section are similar to those by Joep Hamersma, Marc J. van Kreveld, Yushi Uno, and Tom C. van der Zanden [1] given the similar nature of the puzzle discussed here to that analyzed in their paper.

Lemma 3.1. *Given a Hamiltonian cycle, H, and a board B, no matter how the n dominoes start, they can always be moved onto the cycle.*

Proof. Let v_f denote the $t - th$ vertex in the Hamiltonian cycle, H. Let us begin at the first free vertex in the cycle, v_f. There are three cases:

Case 1: If the vertex that follows v_f in the cycle H, namely v_{f+1}, is occupied and the domino is in the cycle, move the domino along the cycle (using a 90° *turn* or a *Straight slide*) and replace v_f with the newly freed vertex. Repeat the process with the next empty vertex in H.

Case 2: If the vertex v_{f+1} is occupied and the domino is not in the cycle move the domino into the cycle (using a 90° *turn* or a *Straight slide*) such that the domino is occupying v_f and the following vertex (which it was already occupying). Repeat the process with the next empty vertex in H.

Case 3: If the vertex v_{f+1} is not occupied, repeat the process with v_{f+1} as the new empty vertex.

Repeat the process until we traverse all of H without encountering *Case 2*.

The algorithm above makes use of Remark 2.5 □

Definition 3.2. *A **U-configuration** refers to a pair of adjacent vertices in a Hamiltonian cycle wherein the cycle makes a turn in the same direction twice, taking on the shape of the letter U.*

Lemma 3.3. *Any Hamiltonian cycle H on a board always has at least one U-configuration.*

Proof. Given the dual graph of a board G_B and a Hamiltonian cycle H of said graph. Let us refer to the square tiling of the polygon defined by H as T_H. The dual graph of such a tiling is then called G_{T_H} (Figure 1). Note that H is a cycle so G_{T_H} must be connected and since B has no holes, G_{T_H} cannot in turn be a cycle. Since G_{T_H} cannot be a cycle, it must have at least one leaf. By construction of G_{T_H} this means that H has at least one U-configuration, namely, the one corresponding to the leaf in G_{T_H}. □

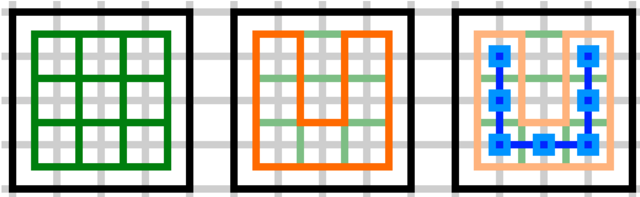

Figure 1: A board B with its board graph G_B (left), a Hamiltonian cycle H in G_B (middle), and the two-leaved dual graph G_{T_H} of the cycle's tiling T_H (right).

Lemma 3.4. *We can always get the two empty spaces in a U-configuration.*

Proof. Choose a Hamiltonian cycle of a given board. Since according to Lemma 3.1, we can always move the dominoes into the cycle, we can assume that all dominoes are already placed on the Hamiltonian cycle. Additionally, from Remark 2.5, we know that a domino adjacent to an empty square can be moved into it. This causes a square previously occupied by the domino to become empty. Let us choose a U-configuration in the cycle (By Lemma 3.3, such a configuration always exists). Using the following algorithm, the two empty spaces can always be moved into said U-configuration:

Start at the initial vertex of the Hamiltonian cycle and cycle through the vertices in the clockwise direction. If an empty vertex, meaning a vertex with no domino in its corresponding square, is encountered, check if that is the first vertex of the given U-configuration (in the clockwise direction). If it is not the first vertex of the given U-configuration move the domino in the previous vertex into the empty vertex according to Remark 2.5. If it is the first vertex of the given U-configuration keep cycling through the vertices until the next free vertex is encountered.

According to remark 2.5 we can cycle the dominoes along the cycle with just one free vertex. By construction of the algorithm described above, we cannot occupy the first vertex of the given U-configuration. Thus, the two empty squares will always end up in the first and second vertex of the given U-configuration. □

Lemma 3.5. *We can always reverse a domino making use of a U-configuration.*

Proof. Given a Hamiltonian cycle, H of a given board, let us choose a U-configuration in H (By Lemma 3.3, such a configuration always exists). Consider the two vertices corresponding to the U-configuration in addition to the vertices in the cycle which are adjacent to each side of the U-configuration. According to Lemma 3.1 the dominoes can always be moved onto H. Thus, assume that the dominoes are on H. According to Lemma 3.4, we can always get two empty spaces in a U-configuration. Thus, we can also assume that the two vertices corresponding to the U-configuration are empty.

Note that given this configuration, we can move one of the dominoes, d, adjacent to the U-configuration into the square corresponding to one of the vertices of the U-configuration (using a 90° *turn* or a *Straight slide*). Then, according to Remark 2.5 we can move the Dominoes along the Hamiltonian cycle using the vertex that was freed when moving d onto the U-configuration. This results in a configuration as can be seen on the left in Figure 2. This configuration allows us to make a *pivot* d thus reversing its orientation with respect to H.

□

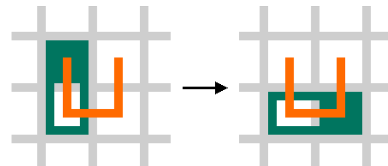

Figure 2: Pivoting of a domino (green) to reverse its order on the Hamiltonian cycle (orange)

Lemma 3.6. *If we consider six squares of the board, namely the two squares corresponding to the U-configuration, and the squares corresponding to the two pairs of vertices before and after the U-configuration in the cycle, and these contain two dominoes and two empty squares, we can always swap the order of the two dominoes in the cycle.*

Proof. Given a Hamiltonian cycle, H of a given board, let us choose a U-configuration in H (By Lemma 3.3, such a configuration always exists). Consider six continuous vertices in a Hamiltonian cycle, H, the vertices corresponding to the chosen U-configuration, the two vertices before the U-configuration, and two vertices after the U-configuration. Assume that all dominoes are on H and that the two vertices corresponding to said U-configuration are empty, according to Lemma 3.1 and Lemma 3.4, no matter how the two dominoes start, we can always obtain such a configuration.

We then have two dominoes occupying four squares corresponding to the vertices adjacent to the U-configuration as shown in the top left in Figure 3. Note that we do not care whether or not there are any turns in H before or after the U-configuration, we do not assume any knowledge regarding the specific layout. Let us refer to the domino occupying the two squares corresponding to the two nodes right before the U-configuration (counterclockwise with respect to H) as d_1, and to the domino occupying the remaining two squares as d_2. (See 3)

Note that if we remove the vertices corresponding to the U-configuration, and connect the end points of the resulting path we obtain H', see Figure 3, which is by the construction of the Hamiltonian cycle corresponding to the board with the squares corresponding to the U-configuration removed. According to Remark 2.5 we can move the dominoes along these H or H' as long as there is at least one vertex corresponding to an empty square.

Thus, in order to swap the order of the two dominoes, we can move d_1 onto the U-configuration (using a 90° *turn* twice or a *Straight slide* and a 90° *turn*). Then, d_2 can then be moved to occupy the two squares previously occupied by d_1 (using a 90° *turn* twice, a 90° *turn* and a *Straight slide* or a *Straight slide* twice). Finally, we can move d_1 along H to occupy the two squares which were initially covered by d_2, thus the order of the two dominoes is swapped. □

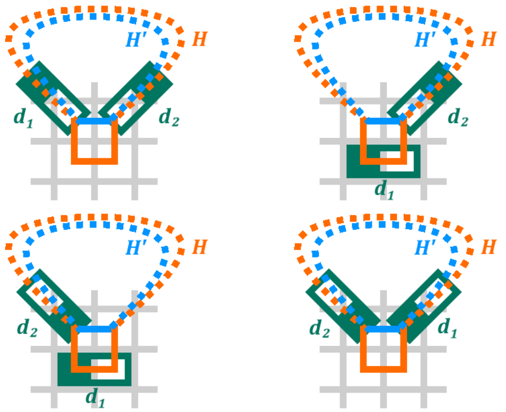

Figure 3: Execution of the domino swap (ordered from left to right, top to bottom).
Note that the Dominoes are drawn diagonally to symbolize that they can be positioned arbitrarily as long as they are aligned with the Hamiltonian cycle H (orange)

THEOREM 3.7. *Any two configurations of dominoes on the board can be transformed into each other.*

PROOF. Let C_0 and C_3 be two arbitrary configurations of the n dominoes on B. Let H be a Hamiltonian cycle of the board B. According to Lemma 3.1 the n dominoes can always be moved onto the cycle. To show that we can transform C_0 into C_3 we begin by moving the dominoes in C_0 onto H. Let us call the resulting configuration C_1. Moreover, let us call C_2 the configuration that results from moving the dominoes in C_3 onto H. We then have two domino configurations C_1 and C_2 where all dominoes are on the same Hamiltonian cycle H. Note that since we can go from C_3 to C_2 we can also go from C_2 to C_3

Hence the only task that remains is to transform C_1 into C_2 which we can do as follows: Choose a U-configuration of H, there must be at least one according to Lemma 3.3. Using said U-configuration, we can reverse the orientation of the dominoes in C_1 such that they are all oriented as in C_2. This is possible according to Lemma 3.5. Now it only remains to reorder the dominoes such that they are in they are also in the right sequence. Assume that there are at least two dominoes, otherwise reordering of the dominoes is trivial. This implies that there must be at least six squares on the board. Consider the squares corresponding to the U-configuration and the squares corresponding to the two vertices before and the two after in the cycle. Assume that the squares corresponding to the U-configuration are empty (according to Lemma 3.4 such a configuration is always possible). We must then have that the remaining four squares are occupied by two dominoes. Then according to Lemma 3.6 we can swap the order of the two dominoes. We can therefore reorder the dominoes in H as needed to obtain C_2. □

4 Further results

We can show that there exists a board that is 2-connected and admits a Hamiltonian cycle, but which does not allow arbitrary reconfigurations (see Figure 4). Note that this does not contradict Theorem 3.7, since such a board has a hole.

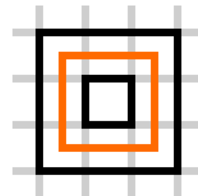

Figure 4: 2-connected board with a Hamiltonian cycle which does not allow arbitrary reconfigurations

By construction, the board in Figure 4 does not allow any pivot moves which implies that it is not possible to reverse a domino.

Furthermore, there exists a 2-connected hole free board which does not admit a Hamiltonian cycle as seen in Figure 5.

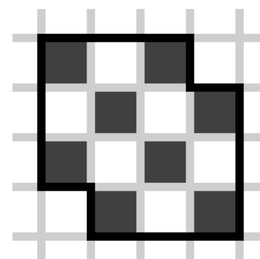

Figure 5: A checkered board with no Hamiltonian cycle

A Hamiltonian cycle must visit every square and return to the initial square. Note that if we colour the board as a checkered board, any path in the board must go from a black square to a white square or vice versa. Moreover, a Hamiltonian cycle must begin and end in the same square so any Hamiltonian cycle will visit as many white squares as it visited black squares. Since the board in Figure 5 has two more black squares than white ones, we can affirm that no such cycle exists.

Verifying that the boards presented in Figure 5 and Figure 4 are 2-connected can be easily done by manually verifying that removing any square of the board does not result in two components.

References

[1] Joep Hamersma, Marc J. van Kreveld, Yushi Uno, and Tom C. van der Zanden. 2020. Gourds: a sliding-block puzzle with turning. *ArXiv* abs/2011.00968 (2020). https://api.semanticscholar.org/CorpusID:226227408